

Building a Water-Quality Data Platform for Future Health Outcome Research

Analyzing Water Quality in Counties Across Texas

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Abstract

Groundwater quality varies substantially across Texas due to heterogeneous geology and aquifer systems. The Texas Water Development Board (TWDB) maintains extensive public groundwater data, including county-level measurements of minerals and elements such as arsenic, fluoride, nitrate, lithium, manganese, and uranium. However, these data are distributed as fragmented text files and spreadsheets, limiting their use for efficient analysis, geospatial visualization, and public health research.

This project developed a normalized MySQL relational database to organize TWDB groundwater-quality data at the county level. We designed a Third Normal Form (3NF) schema, implemented an Entity–Relationship (ER) diagram, cleaned and standardized raw datasets, and created SQL queries to support complex joins and aggregations. The database was then integrated with Python for exploratory mapping and analysis using pandas, GeoPandas, and county shapefiles. Preliminary results demonstrated the ability to summarize mineral concentrations by county, identify geographic patterns, and generate choropleth maps for components of interest. The system provides a scalable foundation for future work linking groundwater quality to health outcomes and for developing advanced spatial and statistical models.

Introduction

Groundwater quality is a key environmental and public health concern across the state of Texas. The Texas Water Development Board (TWDB) maintains one of the most comprehensive groundwater databases in the United States, consisting of decades of mineral and element measurements from wells distributed across all Texas counties (Texas Water Development Board [TWDB], n.d.-a, n.d.-b). These measurements include components such as arsenic, fluoride, nitrate, lithium, manganese, uranium, and hundreds of additional minerals and trace elements. Exposure to degraded drinking water has been associated with a range of adverse health outcomes, including cancer, cardiovascular disease, neurodevelopmental effects, and other chronic conditions (Alzahrani & Alhussain, 2025; Amery et al., 2025; Levin et al., 2024; Srivastava & Malik, 2025; Zhang et al., 2023).

Although TWDB groundwater-quality data are publicly available, they are primarily provided as a large collection of text files, spreadsheets, and static tables with inconsistent formatting (TWDB, n.d.-a, n.d.-b, n.d.-c). As a result, researchers face significant challenges when attempting to answer basic questions such as:

- Which counties have the highest concentration levels?
- Do certain minerals co-occur geographically?
- How do concentrations vary across regions and aquifers over time?

Without a structured relational database, analyses must be performed manually and repeatedly, creating barriers for environmental scientists, policymakers, and data analysts.

The central problem this project addresses is that TWDB’s groundwater-quality data are not stored in an accessible structure that supports efficient querying, aggregation, or integration with analytical tools. The guiding research question for this work is:

How can TWDB groundwater-quality data be transformed into a normalized relational database that enables efficient county-level mineral analysis and supports broader environmental and public health research?

To address this question, we designed and implemented a normalized MySQL database containing county-level mineral composition data. This paper describes the database design, methodology, and early results, and proposes future extensions for large-scale environmental analysis and health-outcome research.

Motivation

Creating a centralized and queryable database enhances the accessibility and transparency of environmental information. Our system supports exploratory data analysis, data quality verification, and visualization in platforms such as Tableau or Python's GeoPandas. This design serves as a scalable foundation for future environmental or geospatial data projects and can be expanded to include additional datasets (e.g., aquifer data or time-series sampling).

Background and Related Work

Water-quality data have been studied extensively across the United States, particularly in relation to public health risks such as cancer, cardiovascular disease, thyroid dysfunction, neurological conditions, and bacterial contamination. Recent work has examined how violations of drinking water standards and chronic exposure to contaminants affect health outcomes at the county level (Alzahrani & Alhussain, 2025; Levin et al., 2024). Specific studies have focused on arsenic exposure and associated health risks, including cancer and cardiovascular outcomes (Amery et al., 2025), as well as multivariate approaches to assessing water pollution and deriving contamination indices (Srivastava & Malik, 2025; Zhang et al., 2023). Research on natural mineral waters has also scrutinized the balance between beneficial mineral content and potential risks from excessive levels of certain constituents (Quattrini et al., 2017; Water Quality Association, n.d.).

Despite the availability of open data portals such as the TWDB Texas Water Data Hub and the EPA Water Quality Portal, few systems provide a relational framework specifically tailored for groundwater-quality analysis at the county level. Most existing platforms offer raw downloads or APIs but lack integrated database-level querying, flexible aggregation, or built-in spatial comparison tools. Because water-quality data involve multiple hierarchical layers, wells, sampling events, mineral parameters, geographic identifiers, and temporal metadata, a normalized relational database offers substantial advantages over raw files for both analytical efficiency and reproducibility.

Our work fills this gap by creating a fully normalized MySQL schema that models the relationships among wells, samples, aquifers, parameters, and counties. This structure supports complex queries and lays the groundwork for higher-level analyses, including environmental justice studies, health-outcome comparisons, and predictive modeling that leverages county-level exposure profiles (e.g., Alzahrani & Alhussain, 2025; Levin et al., 2024).

Methodology

This study aimed to map the distribution of mineral components, such as, but not limited to, arsenic and lead, across Texas counties, identifying areas with elevated concentrations and potential environmental risk. By organizing county-level water-quality data into a relational database, we enabled spatial analysis to highlight regions that may motivate further research into potential related health outcomes.

Our methodology consisted of four major stages: (1) data ingestion and cleaning, (2) relational database design, (3) SQL query development, and (4) Python integration for exploratory mapping and analysis.

Data Collection

This project focused on groundwater-quality data from the Texas Water Development Board's publicly accessible Groundwater Database (GWDB). TWDB's GWDB contains well information, water-quality data, and related hydrogeologic attributes across Texas (TWDB, n.d.-a, n.d.-b). The raw dataset used in this project included (see *Table A1*):

- County name and FIPS code
- State Well ID
- Aquifer name
- Mineral or chemical parameter
- Concentration values
- Units of measure
- Sampling date and time
- Reliability flags and measurement-status codes

The raw files contain hundreds of mineral types and thousands of sampling events spanning more than two decades.

Data Cleaning and Standardization

Prior to loading the data into MySQL, extensive data-cleaning steps were required to resolve inconsistencies in formatting, measurement units, geographic boundaries, and temporal alignment. The main steps included:

- Standardizing units (e.g., mg/L vs. µg/L) to create consistent scales across datasets
- Cleaning and harmonizing parameter names to address variations in naming conventions for the same constituent
- Normalizing date formats to ensure consistent temporal fields for time-based queries
- Resolving null or missing values using appropriate coding or exclusion rules, depending on analytic needs
- Aligning county identifiers across datasets using FIPS codes and TWDB's GIS resources (TWDB, n.d.-c)
- Ensuring consistent geographic information, including coordinate formats for well locations

These steps eliminated many inconsistencies and reduced the risk of referential-integrity issues once the data were imported into the relational schema.

Relational Schema Design

The database was designed in Third Normal Form (3NF) to ensure minimal redundancy and maximum query efficiency. The schema contains 64 tables, including:

- **county** – county identifiers and geographic metadata
- **welldata** – well metadata including aquifer, drilling method, owner, and coordinates
- **waterqualityheader** – metadata for each sampling event (e.g., date, agency, reliability flags)
- **waterqualitydetail** – individual mineral concentration values associated with each sampling event
- **waterqualityparameters** – mineral names, measurement units, and classification fields
- Supporting tables for aquifers, drilling methods, depth sources, and measurement-status flags

Primary keys and foreign keys were used to enforce relationships among wells, sampling events, mineral parameters, and counties. This structure allows users to retrieve complete water-quality histories for any county or well and to aggregate concentration levels across spatial or temporal dimensions (see *Figure A1*).

SQL Query Development

To validate the database design and demonstrate its analytical capabilities, we developed SQL queries that join across multiple levels of the schema. Examples include:

- Counting wells per county
- Listing wells with available water-quality measurements
- Identifying counties with the highest concentration levels for a given mineral
- Computing average mineral concentrations across multiple decades
- Filtering results by mineral type, date range, and reliability flags

These queries confirmed that relational links among the tables were functioning correctly and that the schema supported efficient analytical workflows.

Python Integration and Analysis Plan

To enable visualization and spatial analysis, the MySQL database was connected to Python via MySQL Connector and SQLAlchemy. After extracting SQL results into pandas, we merged them with Texas county shapefiles obtained from TWDB's GIS resources (TWDB, n.d.-c) and produced choropleth maps using GeoPandas and Folium.

The analysis plan centered on:

- Summarizing mineral concentrations by county for selected component (e.g., arsenic, fluoride, nitrate, lithium)
- Visualizing spatial distributions of these components across Texas to identify potential hotspots
- Preparing the database and Python pipeline for future integration with health-outcome datasets and more advanced spatial-statistical analyses, informed by approaches used in previous water-quality and health studies (e.g., Alzahrani & Alhussain, 2025; Levin et al., 2024; Srivastava & Malik, 2025)

Results

The fully operational MySQL database and accompanying analysis pipeline successfully transformed TWDB's raw groundwater-quality files into a structured, analyzable system.

Database Validation

SQL joins demonstrated accurate relational integrity across wells, samples, minerals, and counties. Aggregation queries correctly produced county-level summaries, and parameter codes mapped consistently to their respective mineral definitions. The ER structure supported the intended workflows and eliminated the need for manual filtering or spreadsheet manipulation, addressing the core problem of fragmented data and limited analytical flexibility.

Mineral Concentration Patterns

Preliminary SQL analyses revealed meaningful geographic variation in multiple components (see *Figure A3*):

- **Arsenic** levels were higher in certain West Texas counties, reflecting underlying geologic variation and consistent with prior work documenting spatial heterogeneity in arsenic exposure (Amery et al., 2025; Levin et al., 2024).
- **Fluoride** concentrations exhibited localized clustering in central and western regions, aligning with known patterns of natural mineral content in groundwater (Quattrini et al., 2017; Water Quality Association, n.d.) (see *Figure A2*).
- **Lithium** concentrations varied substantially across counties, suggesting potential geologic influences and offering a starting point for exploring relationships with neurobehavioral outcomes reported in the literature (Levin et al., 2024; Quattrini et al., 2017).

Because TWDB sampling frequency and coverage vary, some rural counties showed sparse data availability. This uneven sampling must be accounted for in future statistical analyses and health-impact assessments.

Python-Based Mapping

The integration of SQL results with geospatial libraries allowed us to produce county-level choropleth maps for several minerals. These maps provided intuitive visualizations of component hotspots and demonstrated how the database can support both exploratory and advanced environmental analytics, including multivariate or PCA-based approaches similar to those used in other water-quality studies (Srivastava & Malik, 2025; Zhang et al., 2023) (see *Figure A2*).

Discussion and Analysis

The creation of a fully normalized relational database significantly improves the usability of groundwater-quality data. Instead of working through fragmented text files and spreadsheets, researchers can now:

- Retrieve complete histories for wells or counties
- Generate county-level component summaries instantly
- Compare multiple minerals simultaneously across regions and aquifers
- Integrate environmental data with geospatial and statistical tools in Python

This system forms the foundation for more sophisticated environmental research, including connections to public health outcomes and environmental justice analyses, in line with emerging work on drinking water quality and health risks (Alzahrani & Alhussain, 2025; Levin et al., 2024).

Strengths

Key strengths of the system include:

- A fully normalized, scalable relational design that supports complex queries
- Clean integration with Python and geospatial libraries, enabling mapping and advanced analytics
- Support for multi-mineral and multi-county analysis, allowing exploration of co-occurring components and spatial variation
- Accurate enforcement of referential integrity, which enhances data reliability and reproducibility

Limitations

Several limitations must be acknowledged:

- Uneven sampling across counties, particularly in rural areas, may bias spatial patterns
- Occasional inconsistencies in measurement units and parameter definitions across years, despite cleaning, warrant cautious interpretation
- Missing data for certain wells or minerals limit the completeness of some time series
- Variation in sampling frequency across decades complicates temporal trend analysis

These limitations highlight the need for further data quality assessment and motivate future enhancements, such as more advanced imputation methods, weighting schemes, or uncertainty analysis similar to approaches used in other water-quality exposure studies (Amery et al., 2025; Srivastava & Malik, 2025).

Conclusion and Future Work

This project successfully developed a relational MySQL database that organizes Texas groundwater-quality data into a format suitable for analytical and spatial exploration. The database structure supports fast SQL-driven analysis, systematic aggregation, and direct integration with Python for visualization. By transforming TWDB's raw files into a relational schema, the system reduces manual effort and creates a reusable data platform for environmental and public health research.

Future directions include:

- Integrating health-outcome datasets from sources such as the CDC, EPA, and NIH to evaluate associations between groundwater components and health indicators (e.g., Alzahrani & Alhussain, 2025; Levin et al., 2024)
- Automating ETL pipelines for both TWDB and health-outcome data to support continuous updates
- Adding temporal analyses to identify trends over multiple decades and to assess the evolution of component profiles (Srivastava & Malik, 2025; Zhang et al., 2023)
- Creating a full interactive Python dashboard for mapping, querying, and generating reports for stakeholders
- Applying predictive models to identify component hotspots or potential risk regions, building on multivariate and machine-learning approaches in the water-quality literature

Ultimately, this system provides a scalable and transparent data platform that can support environmental monitoring, geospatial research, and public health studies across Texas.

References

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Appendix A

Figures and Supplemental Materials

Figure A1

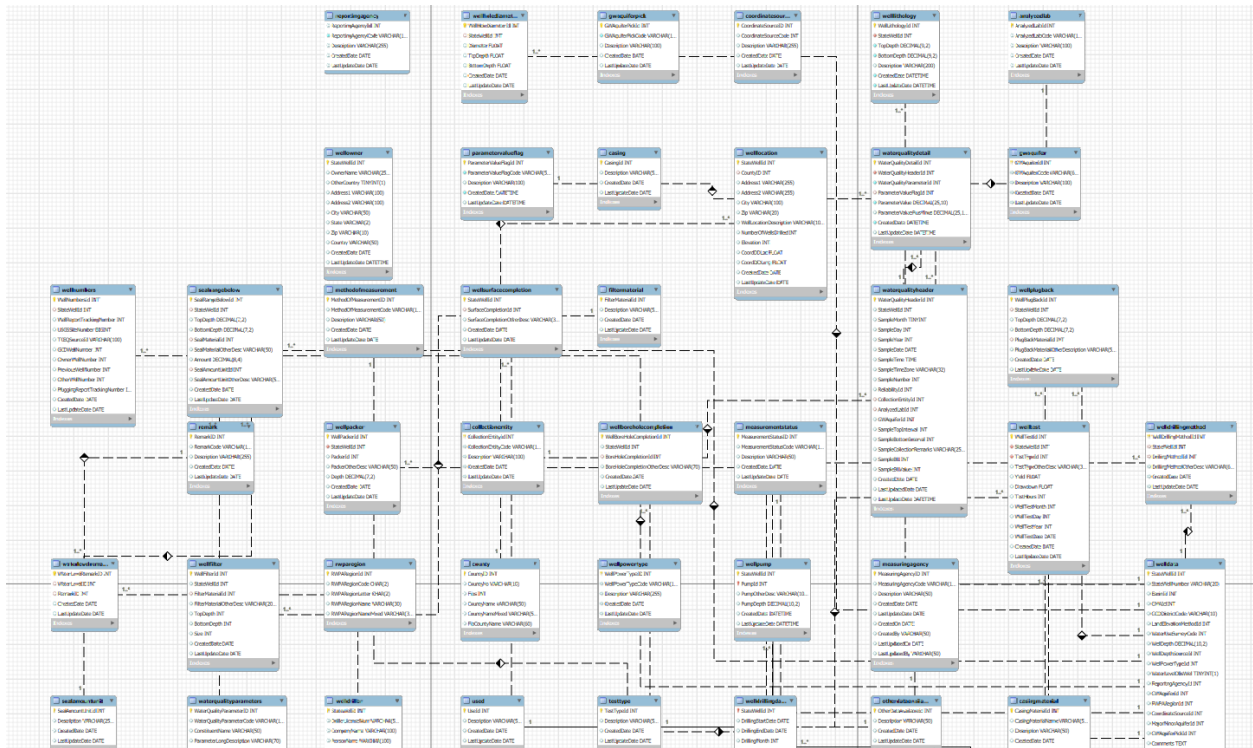


Figure A1

Entity–relationship (ER) diagram for the TWDB groundwater-quality relational database. This diagram displays the core tables—such as *county*, *welldata*, *waterqualityheader*, *waterqualitydetail*, and *waterqualityparameters*—and their primary and foreign key relationships. These structures support county-level aggregation, well-level histories, and multi-mineral analysis.

Figure A2

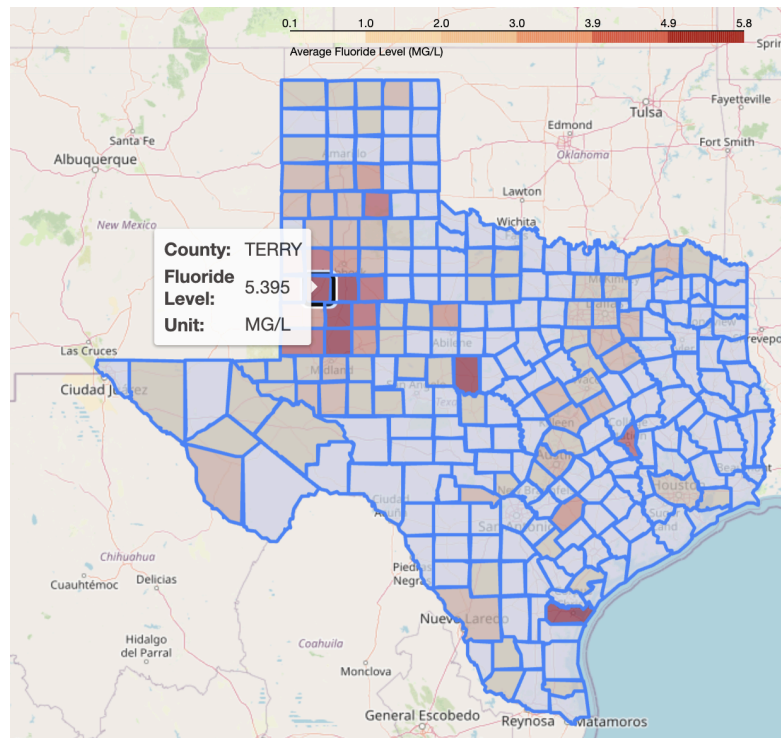


Figure A2

County-level choropleth of mean fluoride concentration (mg/L) in Texas from 2000–2024. Darker shading indicates higher average fluoride levels. This visualization highlights geographic variability and suggests localized regions with elevated natural fluoride concentrations. Data source: Texas Water Development Board (TWDB, n.d.-a; TWDB, n.d.-c).

Figure A3

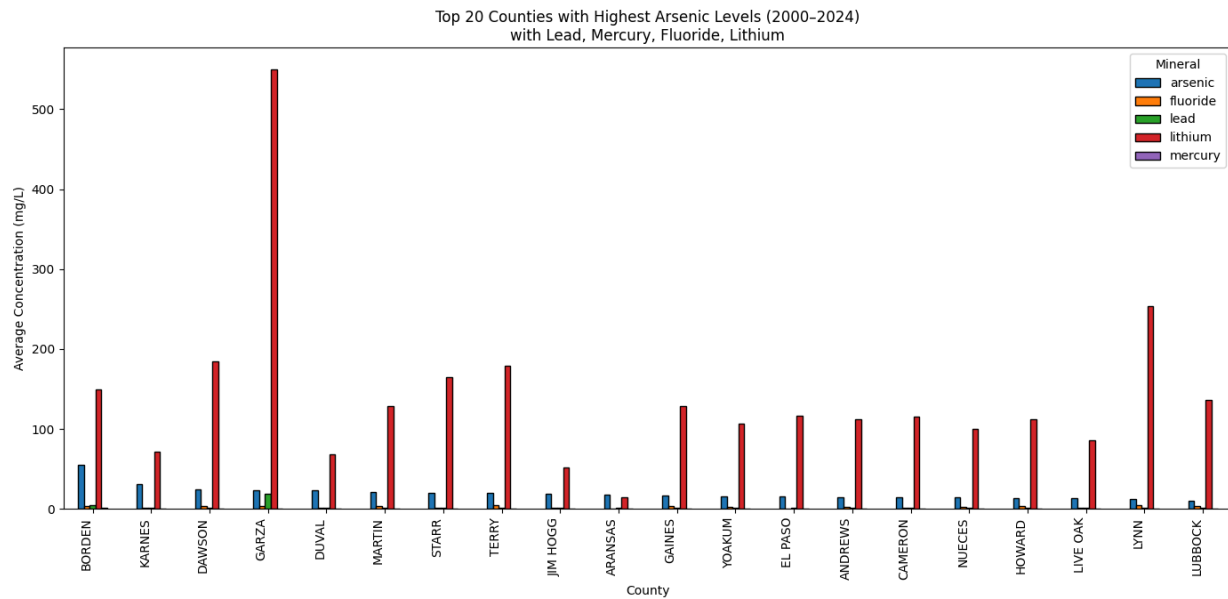


Figure A3

Top 20 Texas counties by average arsenic concentration (2000–2024), with co-occurring components (fluoride, lead, lithium, mercury). Garza County displays the highest average arsenic concentration across the 24-year period. Colors encode different components, as indicated in the legend. The dominance of the red bars illustrates arsenic's relative prevalence compared to other minerals in these counties. This figure is descriptive and was generated in Jupyter Notebook using SQL outputs from the MySQL groundwater database.

Table A1 — Data Dictionary

Field Name	Data Type	Description	Example / Notes
WELL_ID	String / ID	Unique identifier for a well site	“TWDB-12345”
LATITUDE	Float	Latitude of the well site (decimal degrees)	32.7767
LONGITUDE	Float	Longitude of the well site (decimal degrees)	–96.7970
ELEVATION	Float	Elevation of ground surface at well (feet or metres)	450 ft
TOTAL_DEPTH	Float	Total depth of well bore (feet or metres)	350 ft
COMPLETION_DATE	Date	Date when well was completed (construction finished)	2018-06-15
AQUIFER_NAME	String	Name of the aquifer or hydrogeologic unit the well accesses	“Edwards”
WELL_TYPE	String	Classification of well (monitoring, public supply, irrigation, etc.)	“Public Supply”
OWNER	String	Name of the well owner or operator	“City of Dallas”
DRILLER	String	Name of drilling contractor	“Acme Drilling Co.”
WATER_LEVEL_DATE	Date	Date of water-level measurement	2024-04-05
WATER_LEVEL	Float	Depth to water from land surface (feet or metres)	120.5 ft
WATER_QUALITY_PARAM	String	Name of water quality parameter measured	“Total dissolved solids”
WATER_QUALITY_VALUE	Float	Value of that quality parameter	450 mg/L
DATA_SOURCE	String	Source or method of data collection	“TWDB GNDB report”
LAST_UPDATED	Date	Date when the record was last updated	2024-10-01
SHAPE	Geometry	GIS geometry type (Point for wells, Polygon for area)	Point geometry

Table A1

Selected fields from the TWDB groundwater-quality data dictionary used in constructing the relational schema.

Table A2 — Example SQL Query

```

96 • SELECT
97     c.CountyName AS County,
98     wd.StateWellNumber,
99     wqp.ConstituentName AS Aquifer,
100     wqp.WaterQualityParameterCode AS AquiferCode,
101     wd.WellDepth,
102     wh.SampleDate AS Date,
103     wh.SampleTime AS Time,
104     wh.SampleNumber,
105     wh.CollectionEntityId AS EntityId,
106     wh.AnalyzedLabId AS LabId,
107     wh.ReliabilityId,
108     wh.SampleCollectionRemarks AS CollectionRemarks,
109     wl.CoordDDLat AS LatitudeDD,
110     wl.CoordDDLong AS LongitudeDD,
111     wqp.WaterQualityParameterCode AS ParameterCode,
112     wqp.ParameterLongDescription AS ParameterDescription,
113     wqd.ParameterValueFlagId AS ParameterFlag,
114     wqd.ParameterValue,
115     wqp.ParameterUnitOfMeasure AS ParameterUnit,
116     wqd.ParameterValuePlusMinus AS PlusMinus
117 FROM waterqualitydetail wqd
118 JOIN waterqualityheader wh ON wqd.WaterQualityHeaderId = wh.WaterQualityHeaderId
119 JOIN welldata wd ON wh.StateWellId = wd.StateWellId
120 JOIN welllocation wl ON wd.StateWellId = wl.StateWellId
121 JOIN county c ON wl.CountyID = c.CountyID
122 JOIN waterqualityparameters wqp ON wqd.WaterQualityParameterId = wqp.WaterQualityParameterID
123 WHERE c.CountyName = 'Dallas';
124
125

```

Table A2

Example SQL query used to compute county-level average mineral concentrations and validate relational joins.

Table A3 — Example Output from SQL Query

County	StateWellNumber	Aquifer	AquiferCode	WellDepth	Date	Time	SampleNumber	EntityId	LabId	ReliabilityId	CollectionRemarks	LatitudeDD	LongitudeDD	ParameterCode	Parameter
DALLAS	3216601	Specific Conductance	94	439.00	1975-06-03	00:00:00	1	1	1	2	pressure tank	32.8325	-97.0122	94	SPECIFIC C
DALLAS	3216601	pH	400	439.00	1975-06-03	00:00:00	1	1	1	2	pressure tank	32.8325	-97.0122	400	PH (STANDI
DALLAS	3216601	Alkalinity, total	410	439.00	1975-06-03	00:00:00	1	1	1	2	pressure tank	32.8325	-97.0122	410	ALKALINITY
DALLAS	3216601	Alkalinity, Phenolphthalein (total hydroxide + 1/2 c	415	439.00	1975-06-03	00:00:00	1	1	1	2	pressure tank	32.8325	-97.0122	415	ALKALINITY
DALLAS	3216601	Bicarbonate	440	439.00	1975-06-03	00:00:00	1	1	1	2	pressure tank	32.8325	-97.0122	440	BICARBONAT
DALLAS	3216601	Carbonate (CO ₃)	445	439.00	1975-06-03	00:00:00	1	1	1	2	pressure tank	32.8325	-97.0122	445	CARBONAT
DALLAS	3216601	Total Hardness	900	439.00	1975-06-03	00:00:00	1	1	1	2	pressure tank	32.8325	-97.0122	900	HARDNESS,
DALLAS	3216601	CALCIUM	910	439.00	1975-06-03	00:00:00	1	1	1	2	pressure tank	32.8325	-97.0122	910	CALCIUM (
DALLAS	3216601	Magnesium	920	439.00	1975-06-03	00:00:00	1	1	1	2	pressure tank	32.8325	-97.0122	920	MAGNESIUM
DALLAS	3216601	SODIUM	929	439.00	1975-06-03	00:00:00	1	1	1	2	pressure tank	32.8325	-97.0122	929	SODIUM, TI
DALLAS	3216601	Sodium Adsorption Ratio	931	439.00	1975-06-03	00:00:00	1	1	1	2	pressure tank	32.8325	-97.0122	931	SODIUM AD
DALLAS	3216601	Sodium, Percent Total Cations	932	439.00	1975-06-03	00:00:00	1	1	1	2	pressure tank	32.8325	-97.0122	932	SODIUM, C
DALLAS	3216601	CHLORIDE	940	439.00	1975-06-03	00:00:00	1	1	1	2	pressure tank	32.8325	-97.0122	940	CHLORIDE,
DALLAS	3216601	FLUORIDE	945	439.00	1975-06-03	00:00:00	1	1	1	2	pressure tank	32.8325	-97.0122	945	FLUORIDE,

Table A3

Example SQL query output of county-level average mineral concentrations.